Chapter 5
Learning Process and Topics in Engineering Technology

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5.1. Introduction

Process of learning requires two important skills which are comprehension reading and problem solving according to expertise. To be an effective learner you need to develop your skills in these two areas. As you approach learning a topic in your classes, you must organize your learning process and follow it in detail. The learning process is the journey of acquiring, understanding, applying, and integrating new knowledge, skills, attitudes, or values. It's not just about absorbing facts, it's about transformation. Before you start reading a chapter or a section in your textbook, establish a purpose, take a survey and write down questions. This means finding out one single piece of information or concept to comprehend that will enable you to solve problems, do this one paragraph at a time as you read the section or the chapter. Survey the reading, look at the heading, subheading, drawings, diagrams tables, figures, charts and photographs and see if it makes sense to you, try to find a relationship between them as you read. Also, write down any questions that you might have as you go through the reading and see if you can answer them after you done, if not go back and review the stuff you just read.

When reading technical material, always have a paper and pen by your side to write down the summary, questions, sketch graphs and checking calculations. Try not to read quickly, focus on concepts like mathematical formulas, periodically recite the material in your own words and write down anything you don't understand. As a learner you need to do the following after you read your section or chapter, recite load and reread to answer questions. Review the important points to reinforce your long-term memory and keep solving problems related to the topics starting with the easiest.

Being an Engineers requires you to become a problem-solvers. Much of your engineering education, and indeed your engineering career, will center on improving your ability to think both logically and creatively to solve problems. There are many types of problems, science, mathematics, engineering, and technical problems.

Problem-solving is at the heart of engineering, innovation, and even daily life. It's not just a skill, it's a mindset. Here's a structured approach tailored to your regular style of problem solving.

1. Define the Problem Clearly

- Understand the scope, constraints, and root cause.
- Ask: *Is this the real problem or a symptom?*

2. Analyze the Situation

- Gather data, observe patterns, and identify key variables.
- Break it down: What are knowns vs unknowns?

3. Brainstorm Solutions

- Generate diverse ideas—don't judge or filter yet.
- Consider multiple angles (technical, ethical, user impact).

4. Evaluate Options

- Use criteria: feasibility, resources, time, and social implications.
- Apply decision-making models (like SWOT, cost-benefit).

5. Develop and Prototype

- Create a small-scale model or simulation.
- Engage in trial-and-error, embracing feedback and iteration.

6. Implement the Best Solution

- Plan and execute the chosen strategy.
- Assign responsibilities and track progress.

7. Review and Reflect

- Did it work? What could be improved?
- Document lessons learned and prepared for future challenges.

Class activity 1

Reflect on your approach to your classes.

1. Do you operate on the principle that you master the material presented in each class before the next class comes?
2. Do you put off studying until a test is announced and then cram for it?
3. If you are in the second category, are you willing to change? What do you need to do to make that change?

5.2. Topics in Engineering Technology

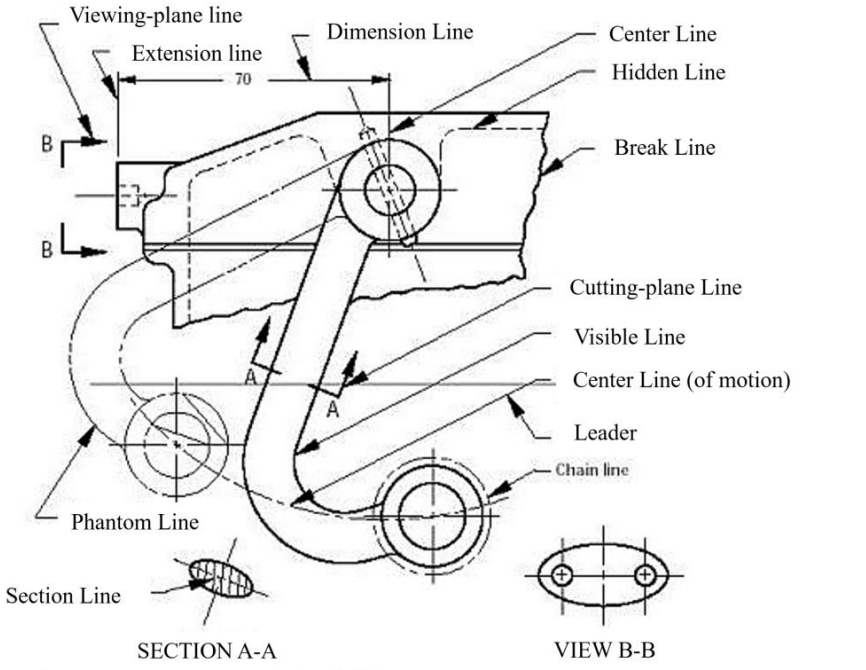
Engineering Drawing Samples

1. **Line Conventions**
    - Visible Lines – solid thick lines that represent visible edges or contours
    - Hidden Lines – short evenly spaced dashes that depict hidden features
    - Section Lines – solid thin lines that indicate cut surfaces
    - Center Lines – alternating long and short dashes
    - Dimensioning
        - Dimension Lines - solid thin lines showing dimension extent/direction

  - Extension Lines - solid thin lines showing point or line to which dimension applies
  - Leaders – direct notes, dimensions, symbols, part numbers, etc. to features on drawing
- Cutting-Plane and Viewing-Plane Lines – indicate location of cutting planes for sectional views and the viewing position for removed partial views
- Break Lines – indicate only portion of object is drawn. May be random "squiggled" line or thin dashes joined by zigzags.
- Phantom Lines – long thin dashes separated by pairs of short dashes indicate alternate positions of moving parts, adjacent position of related parts and repeated detail
- Chain Line – Lines or surfaces with special requirements

2. Lettering

- Plain Gothic
- *Italics are OK*
- ABCDEFGHIJKLMNOPQRSTUVWXYZ
- abcdefghijklmnopqrstuvwxyz



Source: http://www.genium.com/pdf/dmpc.pdf

Figure 1 - diagram of a mechanical engineering scheme showing the different lines of conversions

Sketching drawings are made without mechanical drawings tools such as:

- Free hand
- Ruler
- Simple drawing program

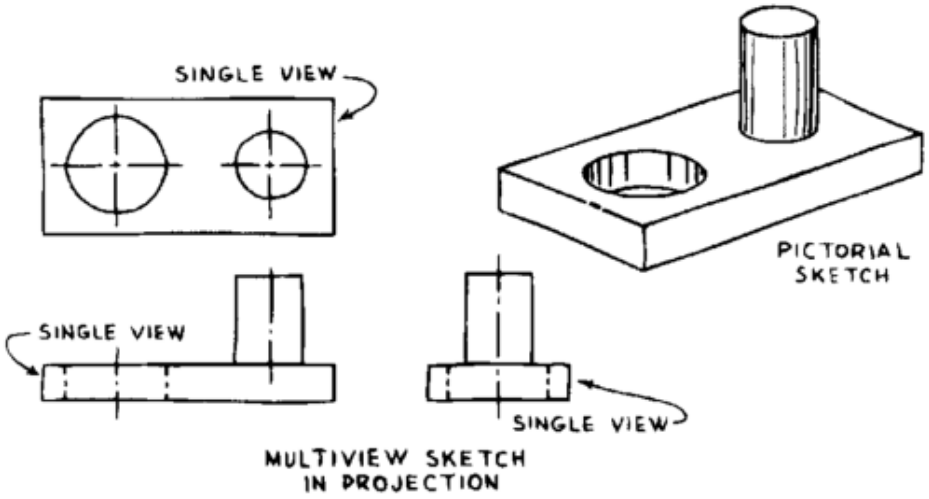


Figure 2 - image of a freehand drawing of a machine piece

Orthographic/Multiview drawings are drawings of objects from two or three perpendicular views.

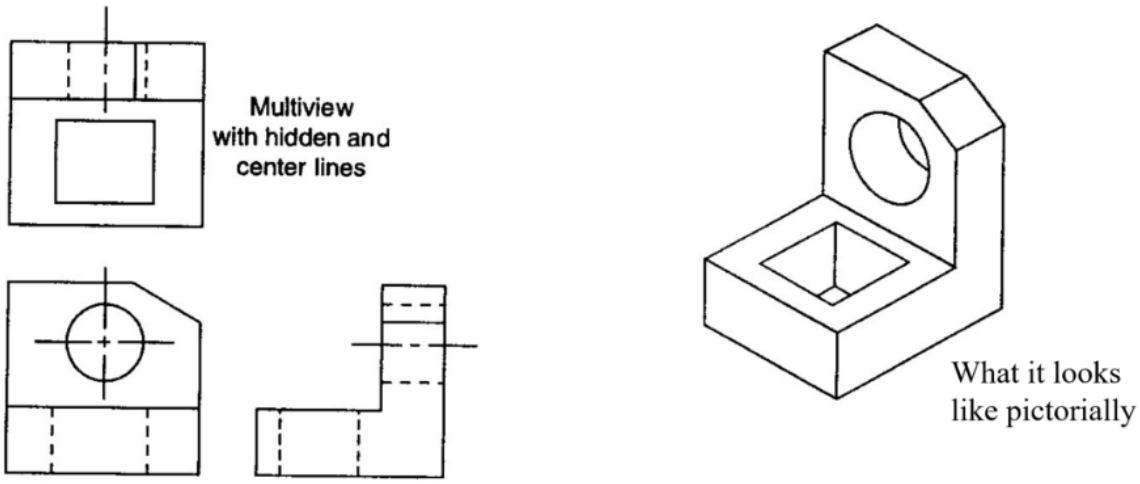


Figure 3 - A diagram of a box drawn in Multiview

Engineering fundamentals – concepts every engineer should know

1. **Length**

   A strong foundation in basic physical quantities is essential for every engineer, as these measurements form the basis of analysis and problem-solving across all engineering disciplines. Core concepts such as length, time, mass, force, temperature, electric current, energy, and power are used daily to design, evaluate, and improve systems. Among these, **length** is one of the most fundamental, playing a critical role in describing physical space, guiding design decisions, and ensuring precision in construction and manufacturing.

   In everyday life, we often use general terms like *tall*, *short*, *long*, *wide*, and *narrow* to describe length and size. These descriptions give a rough idea of dimensions without exact measurements, which may be sufficient for casual conversations. However, in engineering, where accuracy is crucial, such vague terms must be replaced by precise, standardized measurements. Still, recognizing these descriptive terms is important, as they often serve as the initial step in identifying and communicating spatial relationships before moving into detailed calculations.

   Understanding the concept of length enables engineers to define and measure the physical world in a consistent and meaningful way. Whether calculating the dimensions of a soda can, determining the size of a highway sign, or designing the width of hallways and supermarket aisles, accurate length measurements ensure both functionality and safety. Even in everyday products like car seats, careful measurement is required to meet ergonomic and safety standards. In all these cases, mastering the concept of length—starting from basic observation to precise application—is a vital skill that supports effective engineering design and problem-solving.

2. **Basic math**

   Basic mathematical principles such as the Pythagorean Theorem, trigonometry, and algebra play a vital role in engineering technology. These concepts are

fundamental for solving real-world problems involving measurements, design, and system analysis. The Pythagorean Theorem is commonly used to calculate distances and verify structural accuracy in construction and manufacturing. Trigonometry helps determine angles, forces, and component positions in fields such as civil, mechanical, and electrical engineering. Algebra is essential for modeling and solving equations related to currents, loads, and material stresses. By integrating these mathematical tools, engineering technology students can develop the problem-solving skills needed to analyze technical challenges, make accurate calculations, and support decision-making in professional practice.

3. System of units

Understanding different systems of units is essential for engineering technician students, as accurate measurements are critical in all areas of engineering technology. The most widely used system is the International System of Units (SI), which is based on metric units such as meters, kilograms, and seconds. Other systems include the British Gravitational (BG) system, which uses pounds, feet, and seconds, and the United States Customary System, commonly applied in various U.S. industries. Each system has its own set of base units and derived units, and professionals often encounter projects that involve conversions between them. Knowing how to work confidently with these systems—and convert accurately between them—is crucial to prevent costly errors, ensure safety, and maintain consistency in engineering designs, calculations, and communication across international and domestic projects.

Internation System of Units

The **SI**, or **International System of Units** (*Système International d'Unités* in French), is the globally accepted standard system of measurement used in science, engineering, and most industries worldwide. It is based on the metric **system** and is designed to provide a consistent and universal framework for measuring physical quantities.

- Basic units:
    - Length: meter (m)

- Mass: kilogram (kg)
- Time: second (s)
- Electric current: ampere (A)
- Temperature: Kelvin (K)
- Amount of substance: mole (mol)
- Luminous intensity: candela (cd)

Why SI Is Important for Engineering Technician Students:

1. Universality – Used in nearly every country and scientific discipline.
2. Precision and Consistency – Essential for accurate calculations, measurements, and technical communication.
3. Ease of Use – Decimal system makes conversions straightforward (e.g., 1 m = 1000 mm).
4. Standardization – Reduces errors when working across teams, countries, or disciplines.
5. Required in many industries – Especially in automotive, aerospace, electronics, and civil engineering.

Hence, learning and using SI units properly is critical for engineering technicians, as it ensures clear communication, proper documentation, and accurate work in both academic and professional settings.

British gravitation, BG, system of units

The **BG unit system**, or **British Gravitational system**, is a system of units primarily used in classical mechanics. It is based on pounds-force (lbf) as the unit of force, feet (ft) as the unit of length, and seconds (s) as the unit of time.

- Base units:
  - Force: pound-force (lbf)
  - Length: foot (ft)

- Time: second (s)

- Derived units:
  - Mass: In the BG system, mass is not a base unit. Instead, it is derived using Newton's Second Law:

    Mass = force/acceleration = $lbf/ft/s^2$

    This derived unit of mass is called the slug, where:

    1 slug = 1 $lbf.s^2$ / ft ≈ 32.174 lbm

Importance in Engineering: The BG system is less commonly used today but is still relevant in some engineering applications, especially in older U.S.-based mechanical engineering literature and calculations involving force and motion. Engineering technician students need to understand the BG system to interpret legacy data, perform correct unit conversions, and avoid confusion when switching between unit systems such as SI, US Customary, and BG.

United States Customary System (USCS)

The **United States Customary System (USCS)** is a system of measurement commonly used in the United States for everyday activities, commerce, construction, and some engineering fields. It is based on traditional English units, which were inherited from the British Imperial System, but has evolved separately.

- Base units:
  - Length:
    - inch (in)
    - foot (ft)
    - yard (yd)
    - mile (mi)
  - mass:

        - ounce (oz)
        - pound (lb)
        - ton (ton)
    - force: pound-force (lbf)
    - temperature: Fahrenheit

Importance for Engineering Technician Students:

1. Used in U.S. Industries – Many fields like construction, manufacturing, and automotive engineering in the U.S. still use USCS.
2. Understanding Legacy Documents – Older technical drawings and manuals are often in USCS.
3. Conversions Matter – Technicians often need to convert between USCS and SI units, especially when working with international teams or standards.
4. Critical for Compliance – Some government or industrial standards (like ANSI or ASTM) still use USCS measurements.

While the **SI system** is globally dominant, the **US Customary System** remains prevalent in the U.S. Knowing both is essential for flexibility and accuracy in technical work.

Conversion between units

Converting between units is a fundamental skill in engineering and technology, where accurate measurements are critical to the success and safety of a project. Since different systems of measurement—such as the International System of Units (SI), the United States Customary System (USCS), and the British Gravitational (BG) system—are used across industries and regions, engineering technicians must be able to translate values from one system to another with precision. Unit conversion ensures consistency in calculations, allows for clear communication between teams using different measurement standards, and

helps avoid costly errors in design, manufacturing, and construction. Mastery of unit conversion not only improves technical accuracy but also supports collaboration in a global engineering environment.

Example 1: A tourist from Europe is helping a friend measure their backyard in the United States. The length of the backyard is measured as 10 feet. Since the tourist is more familiar with meters, they want to convert the length.

Conversion factor: 1 foot=0.3048 meter

Calculation: 10 ft × conversion factor = 10 ft × 0.3048 m/ft = 3.048 meters

Answer: 10 feet = 3.048 meters

Example 2: An engineering technician in the U.S. receives a shipment of steel parts weighing 5 pounds. They need to record the weight in kilograms to match the project documentation, which follows SI units.

Conversion factor: 1 pound = 0.4536 kilogram

Calculation: 5 lb × conversion factor = 5 lb × 0.4536 kilogram

Answer: 5 lb is 2.268 kg

Class activity 2

Using a ruler, measure the width of three lab components at your workstation. If you don't have access to a workstation, measure three personal items instead, for example, a cellphone, a pen, or a notebook. Record the measurements along with their respective units (e.g., inches or centimeters). Then, using the conversion factors below, convert each measurement into **feet** and **meters**.

Conversion factor:

- 1 inch = 0.0833 feet
- 1 inch = 0.0254 meters
- 1 centimeter = 0.0328 feet
- 1 centimeter = 0.01 meters

For example, if you measured the length of your notebook, then write your answer, with its respective unit, as:

- Notebook in inches =
- Notebook in feet =
- Notebook in meter =

Units of temperature

Temperature conversion is the process of changing a temperature value from one scale to another—commonly between Celsius (°C), Fahrenheit (°F), and Kelvin (K). Since different countries and scientific fields use different temperature scales, converting between them is essential for clear communication, accurate data interpretation, and consistent calculations.

Why Temperature Conversion Is Important:

1. Global Collaboration: Engineers and technicians often work with international teams or reference scientific data reported in different temperature scales. Converting temperatures ensures everyone understands the conditions correctly.
2. Equipment and Process Compatibility: Some machinery, sensors, or processes require temperature inputs in specific units. Accurate conversion prevents operational errors and equipment damage.
3. Safety: In fields like chemical processing or material testing, knowing precise temperatures in the correct units is vital to avoid hazardous situations.
4. Scientific Accuracy: Many formulas and calculations in thermodynamics, fluid mechanics, and material science depend on temperature values in specific units, especially Kelvin for absolute temperature.

Here are the main formulas for converting between Celsius (°C), Fahrenheit (°F), and Kelvin (K):

1. Celsius to Fahrenheit: $°F = (°C \times 9/5) + 32$
2. Celsius to Kelvin: $K = °C + 273.15$
3. Fahrenheit to Celsius: $°C = (°F - 32) \times (5/9)$
4. Fahrenheit to Kelvin: $K = °C + 273.15$
5. Kelvin to Celsius: $°C = K - 273.15$
6. Kelvin to Fahrenheit: $°F = (K - 273.15) \times (9/5) + 32$

Example 3: An engineering intern from Europe checks a thermometer reading in a U.S. lab that shows 98.6°F. They want to know the temperature in Celsius, the system they are familiar with.

Conversion formula: $°C = (°F - 32) \times (5/9)$

Calculation: $°C = (98.6 - 32) \times (5/9) = 37°C$

Answer: $98.6°F = 37°C$

Class activity 3

Find the current temperature in your neighborhood (you may use a weather app or website), and record the temperature in Fahrenheit (°F). After it, convert the temperature to Celsius (°C) and Kelvin (K) using the conversion formulas of temperature.

Write your answers as:

- Temperature in Fahrenheit (°F) =
- Temperature in Celsius (°C) =
- Temperature in Kelvin (K) =

References

chatGPT. (2025). *unit of conversions*. Retrieved from chatGPT: https://chatgpt.com/